

Pump Selection Criteria Refinery

Page 1: Pump Classification and NPSH Sizing

Refineries use centrifugal pumps for high-flow low-viscosity service, and positive displacement pumps for low-flow high-viscosity or metering services. Centrifugal pumps must conform to API 610 standards to ensure mechanical integrity in high-temperature, hydrocarbon services.

Net Positive Suction Head Available (NPSHa) must exceed the Net Positive Suction Head Required (NPSHr) by the pump: $NPSHa = P_{\text{suction_tank}} + H_{\text{static_head}} - P_{\text{vapor_pressure}} - H_{\text{friction_loss}}$. A safety margin of at least 1.0 meter or 10% of NPSHr, whichever is greater, must be maintained to prevent impeller cavitation.